

Gradient function graphs



1

- a 1
- b 1
- c A linear function has a constant gradient.

2

- a The gradient function is positive when $y = x^2$ is increasing, and negative when $y = x^2$ is decreasing.
- b Positive
- c The gradient of the tangent line increases at a constant rate.
- d Negative
- e The gradient of the tangent line increases at a constant rate.
- f Linear

3

- a Positive
- b The gradient of the tangent line increases at an increasing rate.
- c Positive
- d The gradient of the tangent line decreases at an decreasing rate.
- e Quadratic

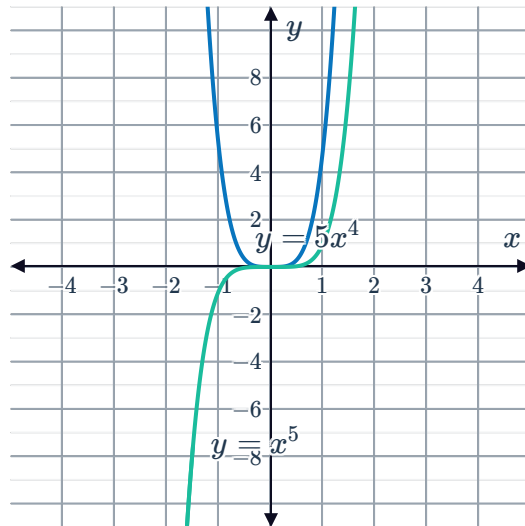
4

- a Positive
- b The gradient of the tangent line increases at an increasing rate.
- c Negative
- d The gradient of the tangent line increases at a decreasing rate.
- e Cubic

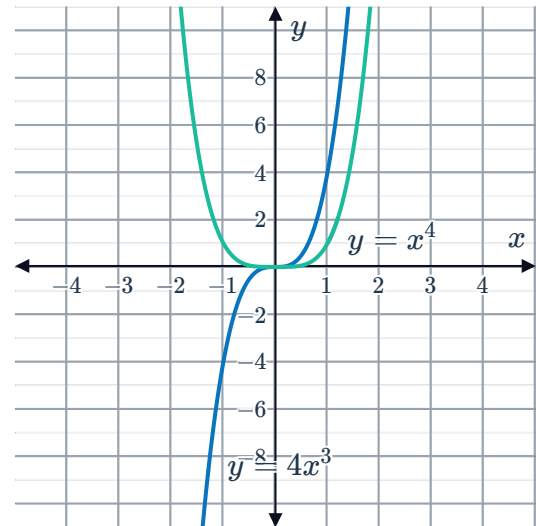
5



a



b



c Odd

d Even

Algebraic differentiation

6

a $2x$

b $8x^7$

c nx^{n-1}

7

a

i Degree 1

ii Degree 2

iii Degree 3

b $n - 1$

8

a $\frac{dy}{dx} = 7x^6$

b $\frac{dy}{dx} = 9x^8$

c $\frac{dy}{dx} = -7x^{-8}$

d $\frac{dy}{dx} = -x^{-2}$

e $\frac{dy}{dx} = \frac{6}{5}x^{\frac{1}{5}}$

f $\frac{dy}{dx} = \frac{2}{3}x^{-\frac{1}{3}}$

g $\frac{dy}{dx} = 0$

h $\frac{dy}{dx} = 0.6x^{-0.4}$

9

a $y = x^{\frac{1}{2}}$

b $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

10

a $y = x^{-2}$

b $\frac{dy}{dx} = -2x^{-3}$

11

a $y = x^{-\frac{1}{4}}$

b $\frac{dy}{dx} = -\frac{1}{4}x^{-\frac{5}{4}}$



12 $\frac{dy}{dt} = \frac{3}{2}t^{\frac{1}{2}}$

13 $\frac{dy}{dx} = -\frac{1}{5x^{\frac{6}{5}}}$

Applications

14 $f'(2) = 32$

15
a 9

b $y = 9x - 8$

16
a $-\frac{1}{2}$

b $x + 2y - 3 = 0$

17
a

Graph	y-intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	2
$y = x^3$	$(0, -2)$	3
$y = x^4$	$(0, -3)$	4
$y = x^5$	$(0, -4)$	5

b n

c No, because then the gradient of the tangent at $x = 1$ would be 1.

18
a False

b True

c False

d True

19
a $\frac{4}{3}x^{\frac{1}{3}}$

b $\frac{3}{4}x^{-\frac{1}{4}}$